■ Message Intelligence System

Supervised Learning — Theory Guide

1. Conditional Probability

Conditional probability is the probability of event A happening when event B is already known to have happened. For example, P(Spam | URL present) is the probability a message is spam given that it contains a URL.

2. Bayes' Theorem

Bayes' theorem updates the probability of a class using observed evidence: P(A | B) = P(B | A) × P(A) / P(B). In classification, it helps estimate the probability that a message belongs to a class after observing its features.

3. Naive Bayes Assumption

Naive Bayes assumes that features are conditionally independent given the class. This makes the model efficient, but correlated features can make the assumption unrealistic.

4. K-Nearest Neighbors (KNN)

KNN finds the K training examples closest to a new example and predicts the majority class among those neighbours. It is distance-based, so scaling is important when features have different units.

5. Support Vector Machine (SVM)

SVM finds a decision boundary that separates classes while aiming for a large margin. A linear kernel uses a straight boundary; an RBF kernel can model non-linear patterns. Support vectors are examples closest to the boundary.

6. Classifier families

KNN is distance-based, Naive Bayes is probabilistic, and SVM is margin-based. They use different assumptions, so evaluating them on the same test set is important.

7. Evaluation metrics

Accuracy is the fraction of correct predictions. Precision measures how many predicted spam messages were truly spam. Recall measures how many actual spam messages were detected. F1 is the harmonic mean of precision and recall.

8. Confusion matrix

True positives are spam correctly identified as spam. False negatives are spam missed by the model. False positives are legitimate messages incorrectly flagged as spam. Both error types matter in a real spam-filtering system.

9. Feature scaling and data splitting

Scaling places numeric features on comparable scales and is important for distance- and margin-based methods. Split data into training and testing sets so final metrics are measured on examples not used to fit the model.

10. Business trade-off

High recall reduces missed spam, while high precision reduces legitimate messages being wrongly flagged. A deployment choice should reflect the cost of each type of error and should be validated on newer, representative data.

Dataset labels: 0 = Legitimate message; 1 = Spam message.